Strengthening accountability for climate change: Experimental evidence on grassroots demand for government action in Mexico

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Abstract

Translating citizen concerns into corresponding action by their governments is a key challenge for democratic institutions. We experimentally study two behavioral science approaches at the research frontier aimed at strengthening citizen pressure on government to combat climate change. The first approach targets social norms describing aggregate levels of concern about, and willingness to act on, climate change. The second approach, "deep engagement," encourages deliberate reflection and system-2 cognition about climate change. 2,000 Mexican adults were randomized into either treatment or a placebo control. The study was conducted online, with initial treatment followed by four short reinforcements spaced at intervals of five days. We measured behaviors, attitudes, and norms through surveys just before treatment and a month after. At endline, both treatments improved dictator-game donations to an anonymous pro-climate citizen, choice of an environmental NGO as recipient of a researcher donation, and willingness to sign a petition asking government to honor its international climate commitments. However, only the social norms treatment rendered participants more optimistic about fellow citizens' concern about, and willingness to act on, climate change, while deep engagement rendered them slightly more pessimistic. Our findings suggest that it is possible to transform citizen concern into grassroots pressure for government climate policy.

Keywords: climate change, environment, grassroots pressure, social norms, deep engagement, accountability, public policy.

Note to SIOE Program Committee: *Thank you for considering this paper. This is work in progress, and a first draft that presents the main elements of the study and the main findings. There is much that we intend to include that is still missing, including a more extensive literature review and a complete references section, additional results, description of prerregistration and IRB authorization, and robustness analyses including balance and differential attrition tests. We also intend to vastly improve formatting and presentation of the results.*

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1. Introduction

The translation of citizen preferences and concerns into corresponding actions by their government is a central challenge of democratic representation. This challenge has become especially pressing in the realm of climate change, where the commitments and actions of governments worldwide remain insufficient to tackle the problem (UNFCCC 2022; Copley 2023; UNEP 2023; WRI 2024) even as the human costs continue to mount. As in other policy areas, public pressure on elected leaders and political parties to combat climate change has the potential to shape policy agendas and actions. Understanding how to strengthen grassroots pressure in the service of combatting climate change constitutes a key frontier for scholars and practitioners.

Theoretical work describes two broad categories of drivers of behavior: nonsocial and social. *Nonsocial* drivers of behavior include knowledge (e.g., about the importance of climate change), preferences (e.g., for a clean environment, or for `doing the right thing'), and beliefs (e.g., about the degree to which one's actions contribute to climate change). Behavior, however, is also importantly driven by what other people do and what they believe is right or wrong. The people whose behaviors and beliefs shape an individual's are termed the individual's `reference group.' Individuals form expectations about how their reference groups are likely to behave. Such expectations are known as `descriptive norms' (Cialdini et al. 1990) or `empirical expectations' (Bicchieri 2017). Individuals additionaly form expectations about how their reference groups judge the appropriateness of specific behaviors—which are known as `injunctive norms' (Cialdini et al. 1990) or `normative expectations' (Bicchieri 2017). We shall refer to behaviors that depend on empirical or normative expectations as `frequency-dependent' or `conditional.' (cites). Thus, by *social* drivers of behaviors we mean factors that underpin frequency-dependent behaviors, specifically empirical and normative expectations or, alternatively, descriptive and injunctive norms.

We experimentally test two interventions aimed at encouraging grassroots political pressure for climate change action. The first intervention, "deep engagement," aims to influence nonsocial drivers of behavior, while the second, "social norms," focuses on social drivers. We focus on interventions that have been identified by behavioral science research as having the potential to attain durable influence on climate change attitudes and behaviors (Goldberg et al. 2020; Vlasceanu et al. 2023). The deep engagement approach encourages deliberate thought about

climate change, its causes and consequences, and the attitudes and actions that could help to perpetuate it or mitigate it. The social norms approach instead targets citizen perceptions about societal levels of belief in, concern about, and willingness to act on, climate change. In other words, it aims to influence descriptive and injunctive norms. A priori, it is not clear whether targeting nonsocial or social factors is best. Insofar as nonsocial factors focus on cognitions and values internalized by the individual, they could exhibit durability independent of what is trendy or fashionable. On the other hand, grassroots political pressure makes itself felt through collective action, where frequency-dependent behavior often plays a central role (cites).[2]

Our study focuses on Mexico—one of the world's biggest emitters of greenhouse gases and, in terms of its political economy and state capacity, a typical middle-income country. As in much of the Global South, the environmental concerns of Mexican citizens often take a back seat to immediate needs stemming from poverty, violence, and corruption. Perhaps for this reason, climate change has not been deeply politicized, as it has been in countries such as the United States. In fact, Mexicans overwhelmingly regard climate change as real and human-made, and view it with great concern (Edelman 2023; Dechelezprêtre et al. 2022). In our study sample, 95% of respondents expressed being worried or very worried about climate change, and similar percentages expressed belief in the reality an in the human origin of climate change. A full 97% expressed feeling a moral obligation to combat climate change through their actions and decisions.

Rather than aiming to convince those who are skeptical about the basic facts of climate change (the focus of a large literature in the United States), our study asks whether it is possible to *activate existing concern about climate change so as to generate social pressure on government to act*—a question at the frontier of behavioral science on climate change (Goldberg et al. 2020, 322). The study aims to contribute to scholarship about persuasion and influence on attitudes and behaviors, both in general and specifically in the environmental realm. Despite the volume and quality of the literature, key questions remain open about how to influence behaviors as opposed to (or in addition to) attitudes, and about how to attain durable influence beyond the moment immediately following an intervention's delivery.

[2] The question of whether social and nonsocial approaches might reinforce each other when applied jointly is beyond the scope of this project.

Our study is designed to contribute to these open questions. In terms of outcome measurement, we utilize material incentives to mitigate bias in behavior variables, to accurately elicit beliefs, and to raise the stakes of participant choices to render them more realistic. We also implement conjoint choice experiments, where marginal effects are difficult to game by any one respondent. We also designed our interventions to have a shot at eliciting durable treatment effects. Measuring outcomes beyond the moment of treatment is important to understand whether a treatment effect is transitory and how it holds up or decays over time. It is also important because measurement biases stemming from experimenter demand effects or social desirability are presumably strongest at the moment of treatment delivery. We therefore measure outcomes both immediately following treatment as well as approximately one month after. The literature additionally suggests that lasting influence is more likely to be attained when treatment exposure is repeated over time, rather than delivered just once. For this reason, our design includes up to four weekly treatment reinforcements after treatment delivery.

The study was conducted over the course of February and March, 2024. Respondents were drawn from a convenience sample of Mexican adults provided by Netquest, an international survey company with a pool of over 100,000 potential panelists in Mexico. The study was conducted entirely online. We randomized participants into a social norms, a deep engagement, or a placebo control treatment condition, and collected outcome measures both immediately after treatment (baseline) as well approximately a month after (endline). The baseline and endline surveys lasted about 30 minutes. We additionally invited respondents to take part in up to four short treatment reinforcement surveys, lasting about 3 minutes each, in the weeks after baseline and before endline. To implement incentivized outcome measures and to mitigate attrition, we utilized cash points redeemable at an online store selling a wide variety of goods including electronics, movie tickets, and household items among other things. Our baseline sample yielded 2,719 complete responses after discarding those who failed a simple attention check. All subsequent participants were drawn from this pool of baseline respondents. The endline sample consists of 2,011 respondents (to stay within budget, we stopped endline data collection just after reaching the target of 2,000 endline completes, as planned from the outset). Our study samples are slightly older and better educated than the average Mexican adult. Nevertheless, randomization of treatment assignment within our sample ensures statistical comparability across treatment arms.

We find that both the social norms and the deep engagement treatments strengthened pro-environment behaviors at baseline and at endline, in comparison with the placebo control. Participants in the two treatment conditions donated more cash points, in a dictator game (to be described in detail further below), to an anonymous study participant revealed to be committed to the environment. They were also more likely to donate to a reforestation NGO, and more willing to sign a petition calling on the Mexican government to address climate change.

Nevertheless, the effects of deep engagement and social norms were not always the same. We find that dictator game allocations of cash points to an anonymous participant said to *not* be committed to the environment were increased by the social norms treatment but not by the deep engagement treatment, compared to control. In addition, applying a simple model of collective behavior (Granovetter 1978), our data suggest that the social norms treatment would increase the societal equilibrium rate of grassroots pressure for climate change (specifically, voting and protesting), while deep engagement would not.

These patterns suggest that the social norms and deep engagement treatments operated differently. The social norms treatment rendered participants more "environmentally optimistic," meaning that it positively shifted their perceptions about the degree of environmentalism (which we use loosely to refer to pro-environment beliefs, preferences, and behaviors) of fellow Mexican citizens and fellow study participants, in comparison with the control. By so doing, it increased willingness to behave pro-environmentally for any given level of societal environmentalism. The deep engagement treatment, in contrast, rendered participants slightly more pessimistic about others' environmentalism, leaving their tendency to participate in grassroots climate action collaboratively, as well as their willingness to donate to a non-environmentalist peer, essentially unchanged with respect to the control.

Taken together, our findings cast both the social norms and the deep engagement interventions as effective means to leverage citizen concern about climate change into citizen action. Both treatments influenced participant preferences in a pro-environment direction, and increased willingness to part with resources to reward an environmentalist peer or NGO, as well as to sign a climate-related petition. Nevertheless, the social norms treatment in addition appeared capable of leveraging the frequency-dependent nature of climate behaviors such as voting and protesting, enhancing the potential for societal coordination on pro-environment collective action. Such potential is evidenced by the elicitation of frequency-dependent willingness to vote

or protest, as well as by direct evidence on perceived descriptive and injunctive norms—both of which were positively influenced by the social norms treatment but left unchanged by the deep engagement approach. A key lesson from these results is that interventions aimed at encouraging behaviors with an important frequency-dependent component ought to be evaluated not only individually, but also in terms of their potential to encourage collective action.

2. Treatment interventions

Participants were randomized into one of three experimental conditions: control, social norms (treatment 1), and deep engagement (treatment 2). The social norms and deep engagement approaches are identified by Goldberg et al. (2020) as promising, evidence-based avenues to "generate lasting engagement with climate change solutions." Our implementation of these approaches draws on the findings of Vlasceanu et al. (2023), an evaluation of behavioral science climate-change interventions in approx. 60,000 individuals in 63 countries. Our treatments are constructed as follows:

Social norms treatment

The social norms treatment consists of exposing the participant to true societal levels of belief in human-made climate change (descriptive norms / empirical expectations), to normative societal evaluations about the importance of acting about climate change (injunctive norms / normative expectations), and to trends in such norms. This intervention draws from work on dynamic social norms (Sparkman & Walton 2017), pluralistic ignorance (Shamir and Shamir 1997; Geiger & Swim 2016), effective collective action (Goldberger et al. 2018; Lizzio-Wilson et al. 2021), and scientific consensus (van der Linden et al. 2015; Rode et al., 2021; Većkalov et al. 2024).

- Goal: to influence participants' perceptions of:
    - Belief in, and willingness to act on, climate change among Mexican adults (descriptive norms and empirical expectations);
    - Existence and content of scientific consensus about climate change (normative belief among authorities);
    - Possibilities for effective collective action on climate change (presumably reflecting collectively-shared expectations).
- Description: The treatment combines the following empirically-tested techniques:
    - Descriptive and injunctive social norms (Wolsko et al. 2016; Sparkman and Walton 2017; Howe et al. 2021)
    - Scientific consensus (van der Linden et al. 2015; Rode et al. 2021; Većkalov et al. 2024):
    - Effective collective action (Goldenberg et al. 2018; Lizzio-Wilson et al. 2021)
    - Pluralistic ignorance (Geiger and Swim 2016)

Deep engagement treatment

The deep engagement treatment encourages deliberate, conscious ("system 2") engagement with the reality of climate change, its consequences, and one's behavior. Our intervention draws on research about future-self continuity (Hershfield et al. 2012), perspective taking (Goldberg et al. 2020; Koessler et al. 2023), psychological distance (Jones et al. 2017), and consideration of future generations (Shrum, 2021; Wickersham, et al. 2020).

- Goal: to influence participants':
    - Conscious and deliberate engagement with the personal costs of climate change
    - Emotional awareness of the downsides of climate change
    - Conscious and deliberate reflection about one's own action or inaction about climate change
- Description: The treatment combines the following empirically-tested techniques:
    - Future-self continuity (Hershfield et al. 2012)
    - Psychological distance (Jones et al. 2017)
    - Negative emotions (Chapman et al. 2017)

Placebo control

- Goal: to occupy participant time during the survey in a way that does not intersect with climate change.
- Description: In the baseline, the participant is asked to identify historical and contemporary famous Mexicans in a series of multiple-choice questions. Followup placebo "reinforcements" were also administered in parallel with the treatment reinforcements. In the first installment, for example, the participant was asked to identify the location of different states of Mexico in unlabeled maps. [A full description of the placebo "reinforcements" will be provided in a future iteration of the manuscript].

3. Study design

Logistics

The study consisted in the following stages, all delivered online and implemented on Qualtrics.

1. Baseline (30 mins). Target N = 2,700 completes.
2. Followup treatment reinforcements: four followup surveys (aprox. 3 mins each) fielded sequentially over the course of 2 weeks to all those who completed stage 1.
3. Endline (aprox. 30 mins): Target N = 2,000 completes, all drawn from baseline completes. Fielded aprox. 3-4 weeks after baseline.

Deployment of the baseline survey began on February 12, 2024 and deployment of the endline survey ended on March 13, 2024. The baseline survey yielded 2,719 complete sets of responses after dropping individuals who did not pass a simple attention check. These respondents were all invited to the followup treatment reinforcements and, regardless of how many of these they completed, they were also invited to respond to the endline survey. Once the number of complete endline respondents reached slightly more than the targeted 2,000 responses, endline data collection was stopped and the study was considered complete. After discarding responses that failed the attention check, the endline survey yielded 2,011 complete responses.

Outcome variables

The following outcome variables were collected at baseline and endline. The full surveys (in the original Spanish) are included in the Appendix.

Incentivized behaviors

These outcomes are incentivized with cash-points redeemable by all participants for goods and services at an online store, including electronics, household items, movie tickets, and others.

1. Dictator allocation to a sympathizer with climate change policy
2. Dictator allocation to a non-sympathizer with climate change policy

3. Donation choice: choose target of donation to be made by research team between: (a) a reforestation NGO, and (b) the study participants
4. Willingness to initial a petition calling for the Mexican government to honor international climate commitments

Political climate perceptions and preferences

5. Conjoint candidate-choice experiments (to estimate the relative importance of different candidate attributes and levels, including commitment to the environment)
6. Perceived commitment of the major Mexican political parties with climate change

Climate attitudes and behaviors

7. Willingness to adopt climate-friendly behaviors
8. Own belief in climate change, human made climate change, and scientific consensus
9. Own worry about climate change
10. Perceived effects of climate change on own welfare (income, family), other social groups (the poor, future generations, city, country), and plants and animals

Climate policy evaluations

11. Conjoint policy-choice experiments (to estimate the relative importance of different policy attributes and levels in policy preferences)
12. Direct evaluations (perceived effectiveness, support, and perceived effects on own household)

Climate-change norms

13. Incentivized elicitation of perceived empirical and normative expectations
14. Nonincentivized elicitation of own behavior conditional on descriptive norms/empirical expectations
15. Perceived societal belief in climate change, human made climate change, and scientific consensus

General political preferences

16. Interest in politics
17. Party feeling thermometers (affective polarization)

To mitigate statistical issues with multiple hypothesis testing, a set of indices was constructed. [The index construction and analysis will be discussed and presented in a future iteration of this manuscript; the conjoint choice experiments and related findings will be included in a future iteration of this manuscript].

Estimation

Estimation of causal effects is straightforward, as participants were randomly assigned to the social norms, deep engagement, and placebo control arms. Throughout, we report differences of means across treatment branches, which constitute unbiased estimates of intent-to-treat effects. [A future iteration of this manuscript will also include regression-based estimates]. Power calculations for these estimates are provided in the Appendix. To check the randomization, we verified balance on pretreatment covariates across treatment arms at baseline (N = 2,719) and at endline (N = 2,011). These tables are provided in the Appendix.

4. Results

This section discusses the results of the estimation exercise, focusing on the results at endline [results at baseline are similar and generally stronger; a future iteration of the manuscript will include and discuss the full set of results including baseline and endline, as well as compliance with followup treatment reinforcements; also, I apologize for the “ugly” tables, they will be better formatted in a future iteration]. In the text below we discuss effect sizes mostly in units of percentage points of the control mean. The tables, however, provide marginal effects in the raw units of measurement. All percentage point magnitudes reported below can be computed directly from the information provided in the tables.

We find that assignment to both the social norms and the deep engagement treatments strengthened environmental behaviors at baseline (i.e., immediately after treatment) and at endline (i.e., approximately one month after initial exposure to the treatment, and following all treatment reinforcements) in comparison with the control. Specifically, as shown in Table 1, participants in the two treatment conditions donated a greater amount of cash points (13 percentage points—henceforth “pp”---of control mean at endline) to an anonymous study participant known to be committed to the environment ($p < .05$), they were more likely to choose a climate-related NGO vs. other study participants as the recipient of a cash donation made by the research team (10 pp of the control mean at endline; $p < .05$), and they were more willing to sign a petition calling for the government to fight climate change (4 to 6 pp of the control mean at endline; $p < .05$). Consistent with the effects on incentivized behaviors, participants in both the social norms and the deep engagement treatments ranked the environment higher than those in the control arm within a list of policy priorities that also included security, poverty, sports, and culture (11 pp higher than in the control condition at endline; $p < .05$) (Table 2).

Table 1

Incentivized climate-related behaviors

Variable	(1) Control N	(1) Control Mean/(SE)	(2) Deep engagement N	(2) Deep engagement Mean/(SE)	(3) Norms N	(3) Norms Mean/(SE)	(1)-(2) N	(1)-(2) Mean difference	(1)-(3) Pairwise t-test N	(1)-(3) Pairwise t-test Mean difference	(2)-(3) N	(2)-(3) Mean difference
Dict. donation to environmentalist	671	127.848 (4.641)	653	145.343 (4.906)	687	145.146 (4.999)	1324	-17.495***	1358	-17.298**	1340	0.197
Dict. donation to non-environmentalist	671	56.900 (3.447)	653	49.340 (3.679)	687	64.119 (3.895)	1324	7.560	1358	-7.219	1340	-14.779***
Choice of donation target (0=participants, 1=NGO)	671	0.565 (0.019)	653	0.625 (0.019)	687	0.622 (0.019)	1324	-0.060**	1358	-0.057**	1340	0.003
Willing to sign climate petition	671	0.849 (0.014)	653	0.893 (0.012)	687	0.895 (0.012)	1324	-0.043**	1358	-0.046**	1340	-0.002
Climate behavior index	909	0.700 (0.011)	898	0.754 (0.010)	918	0.725 (0.010)	1807	-0.053***	1827	-0.025*	1816	0.028**

Notes: Based on complete responses at endline who passed the attention check (N = 2,011). * p<.1; ** p<.05; *** p<.01.

Table 2

Participant ranking of priorities for government policy

Variable	(1) Control N	(1) Control Mean/(SE)	(2) Deep engagement N	(2) Deep engagement Mean/(SE)	(3) Norms N	(3) Norms Mean/(SE)	(1)-(2) N	(1)-(2) Mean difference	(1)-(3) Pairwise t-test N	(1)-(3) Pairwise t-test Mean difference	(2)-(3) N	(2)-(3) Mean difference
Environment	671	2.638 (0.038)	653	2.357 (0.037)	687	2.342 (0.038)	1324	0.281***	1358	0.296***	1340	0.015
Poverty	671	2.298 (0.040)	653	2.361 (0.040)	687	2.437 (0.039)	1324	-0.063	1358	-0.139**	1340	-0.075
Security	671	1.560 (0.031)	653	1.674 (0.034)	687	1.654 (0.033)	1324	-0.113**	1358	-0.093**	1340	0.020
Sports	671	4.535 (0.029)	653	4.570 (0.027)	687	4.543 (0.028)	1324	-0.035	1358	-0.008	1340	0.027
Culture	671	3.969 (0.034)	653	4.038 (0.033)	687	4.025 (0.032)	1324	-0.070	1358	-0.056	1340	0.014

Notes: Based on complete responses at endline who passed the attention check (N = 2,011). * p<.1; ** p<.05; *** p<.01.

At the same time, there were important differences between the effects of the social norms vs. the deep engagement conditions. While, as mentioned above, both the social norms and the deep engagement treatments increased cash point donations to the environmentally committed, they had opposite effects on donations to an anonymous participant *not* committed to the environment. Assignment to the social norms condition increased donations with respect to the control, while deep engagement decreased them (Table 1). While neither effect was statistically significant in comparison with the control, they were statistically different from each other (the difference was 30 percentage points of the deep engagement mean donation at endline; $p < .05$).

One potential explanation for the differential effect of the social norms and the deep engagement treatments is that the social norms treatment substantially changed perceptions of the average study participant and the average fellow citizen, while the deep engagement and control conditions did not. As shown in Table 3, participants assigned to social norms viewed fellow Mexican citizens as more likely to regard the environment as a major problem (31 percentage points of the control mean at endline; $p < .01$) and to believe that climate change is caused by humans (18 percentage points of the control mean at endline; $p < .01$). They also expected fellow Mexicans to behave more pro-environmentally: more likely to sign a petition calling for government to fight climate change (23 percentage points of control mean at endline; $p < .01$), more likely to consider environmental issues when voting in elections (22 pp of control mean at endline; $p < .01$), and more likely to participate in a peaceful climate-change protest (22 pp of control mean at endline; $p < .01$). In other words, the social norms treatment importantly influenced "descriptive norms" or "empirical expectations," understood as individual beliefs about the likely behavior of others (Cialdini 1990; Bicchieri 2017). [A finer-grained discussion of effects on social norms will be provided in a future iteration of the manuscript]

Table 3

Participant beliefs about fellow Mexican citizens

	(1) Control		(2) Deep engagement		(3) Norms		(1)-(2)		(1)-(3) Pairwise t-test		(2)-(3)	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference	N	Mean difference	N	Mean difference
Climate change real	671	5.091	653	4.900	687	6.667	1324	0.190	1358	-1.576***	1340	-1.766***
		(0.092)		(0.093)		(0.081)						
Climate change human-caused	671	6.161	653	5.819	687	7.272	1324	0.342**	1358	-1.111***	1340	-1.453***
		(0.102)		(0.105)		(0.081)						
Vote environmentally	671	4.633	653	4.435	687	5.662	1324	0.198	1358	-1.029***	1340	-1.227***
		(0.097)		(0.100)		(0.089)						
Participate in climate protest	671	4.727	653	4.715	687	5.748	1324	0.012	1358	-1.021***	1340	-1.033***
		(0.094)		(0.100)		(0.088)						
Sign climate change petition	671	5.109	653	5.182	687	6.293	1324	-0.073	1358	-1.184***	1340	-1.110***
		(0.099)		(0.102)		(0.092)						

Notes: Based on complete responses at endline who passed the attention check (N = 2,011). * p<.1; ** p<.05; *** p<.01.

To elicit the perceived environmentalism of fellow study participants, we incentivized accuracy by offering cash points to those who correctly guessed (a) the fraction of study participants who regard themselves as making an effort to combat climate change, and (b) the fraction of study participants who feel a moral obligation to fight climate change through their actions and choices. The results are shown in Table 4. While all participants substantially underestimated

the true fractions—suggesting an important degree of pluralistic ignorance—guesses by those assigned to the social norms conditions were on average much higher than guesss by those in the deep engagement or control conditions (17 to 20 pp of the control mean at endline; p < .01). Summing up, one model consistent with the findings is that the social norms treatment shifted distribution of perceived environmentalism of fellow citizens and fellow study participants to the right, and thereby increased willingness to donate cash points to study participants of all descriptions (i.e., irrespective of the stated commitment of such participants to the environment).

Table 4

Participant beliefs about fellow study participants

	(1) Control		(2) Deep engagement		(3) Norms		(1)-(2)		(1)-(3) Pairwise t-test		(2)-(3)	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference	N	Mean difference	N	Mean difference
Make effort against climate chg.	671	0.832	653	0.871	687	0.885	1324	-0.040**	1358	-0.053***	1340	-0.014
		(0.014)		(0.013)		(0.012)						
Moral obligation to act against climate chg.	671	0.946	653	0.966	687	0.962	1324	-0.020*	1358	-0.016	1340	0.004
		(0.009)		(0.007)		(0.007)						
Guess about climate effort	671	61.887	653	60.268	687	72.431	1324	1.619	1358	-10.544***	1340	-12.163***
		(1.022)		(0.988)		(0.806)						
Guess about climate moral obligation	671	62.095	653	63.089	687	74.173	1324	-0.993	1358	-12.078***	1340	-11.084***
		(1.055)		(1.050)		(0.823)						

Notes: Based on complete responses at endline who passed the attention check (N = 2,011). * p<.1; ** p<.05; *** p<.01.

The findings also document differential effects of the social norms vs. deep engagement treatments on self-reported behaviors (Table 5). Participants assigned to the social norms arm reported walking and biking more, talking about the environment with friends and family more frequently, than those assigned to deep engagement or the control (11 and 8 percentage points of the control mean at endline respectively; p < .05 and p < .1 respectively), and watching news more frequently (5 pp of the control mean at endline; p < .1).

Table 5

Nonincentivized, self-reported climate-related behaviors

Variable	N	(1) Control Mean/(SE)	N	(2) Deep engagement Mean/(SE)	N	(3) Norms Mean/(SE)	N	(1)-(2) Mean difference	N	(1)-(3) Pairwise t-test Mean difference	N	(2)-(3) Mean difference
Time in public transportation (hrs/day)	525	1.366 (0.052)	479	1.382 (0.057)	517	1.461 (0.055)	1004	-0.016	1042	-0.095	996	-0.079
Time walking or biking (hrs/day)	513	1.416 (0.055)	520	1.523 (0.057)	532	1.576 (0.059)	1033	-0.107	1045	-0.160**	1052	-0.053
Time in car/taxi/motorcycle (hrs/day)	529	1.384 (0.057)	519	1.382 (0.058)	572	1.325 (0.054)	1048	0.003	1101	0.060	1091	0.057
Talk about environment w fam. & friends	671	1.477 (0.047)	653	1.567 (0.049)	687	1.598 (0.046)	1324	-0.090	1358	-0.121*	1340	-0.032
Talk about politics w fam. & friends	671	1.382 (0.052)	653	1.354 (0.053)	687	1.320 (0.048)	1324	0.028	1358	0.061	1340	0.034
Freq. of news watching	671	2.787 (0.056)	653	2.737 (0.057)	687	2.916 (0.051)	1324	0.050	1358	-0.129*	1340	-0.179**

Notes: Based on complete responses at endline who passed the attention check (N = 2,011). * p<.1; ** p<.05; *** p<.01.

There were also differential effects of the treatments on attitudes, as shown in Table 6. The social norms treatment strengthened perceptions that climate change is real (3 pp of the control mean at endline; $p < .05$) and human-caused (2 pp of control mean at endline; $p < .1$), and it enhanced trust in climate science and scientists (5 pp of the control mean at endline; $p < .01$), while deep engagement was indistinguishable from the control.

Table 6

Participant own beliefs about climate change

Variable	N	(1) Control Mean/(SE)	N	(2) Deep engagement Mean/(SE)	N	(3) Norms Mean/(SE)	N	(1)-(2) Mean difference	N	(1)-(3) Pairwise t-test Mean difference	N	(2)-(3) Mean difference
Climate chg. real	671	5.361 (0.046)	653	5.441 (0.045)	687	5.511 (0.041)	1324	-0.080	1358	-0.150**	1340	-0.070
Climate chg. human made	671	5.276 (0.043)	653	5.308 (0.041)	687	5.376 (0.042)	1324	-0.032	1358	-0.100*	1340	-0.068
Trust in climate science/scientists	671	4.539 (0.050)	653	4.614 (0.048)	687	4.779 (0.048)	1324	-0.075	1358	-0.239***	1340	-0.165**

Notes: Based on complete responses at endline who passed the attention check (N = 2,011). * p<.1; ** p<.05; *** p<.01.

Overall, the pattern that emerges is that both social norms and deep engagement yielded pro-environment effects in comparison with the control, but social norms appears to have been more effective than deep engagement. There were no instances in which deep engagement caused more pro-environment attitudes or behaviors than the social norms treatment. The social norms treatment increased participants' optimism about others' environmental attitudes and about their likelihood of behaving in a pro-environment manner. The deep engagement

treatment, in contrast, appears to have operated through a different mechanism—potentially by inducing "negativity." Those assigned to deep engagement, for example, came to view fellow Mexicans as *less* trusting that climate is caused by humans (6 pp of the control mean at endline, $p < .05$; Table 3) and reported a *decrease* in the frequency of news consumption (7 pp lower than the social norms treatment mean; $p < .05$; Table 5)---both effect of opposite sign to those of the social norms treatment.

5. Conclusions

Our findings contribute to the question of "how to create sustained engagement with the issue of climate change," identified as a research frontier in behavioral science (Goldberg et al. 2020). Because the Mexican population is already highly convinced that climate change is real, human made, and worrisome, our findings speak more directly to the question of how to activate the already-engaged than how to persuade the uninformed or the skeptic that a problem exists in the first place.

Future research might test effect durability in longer time periods, as well as the relationship between the quantity and periodicity of treatment reinforcements, on the one hand, and treatment effects and durability on the other. Because each of our treatments bundles various subtypes of interventions, we are unable to attribute treatment effects to specific elements within each bundle. Future work might aim to better pinpoint the aspects of treatments most responsible for the measured causal effects. Finally, the consequences of the differences in the manner in which our social norms vs. deep engagement interventions operated remain an open question ripe for further investigation.

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